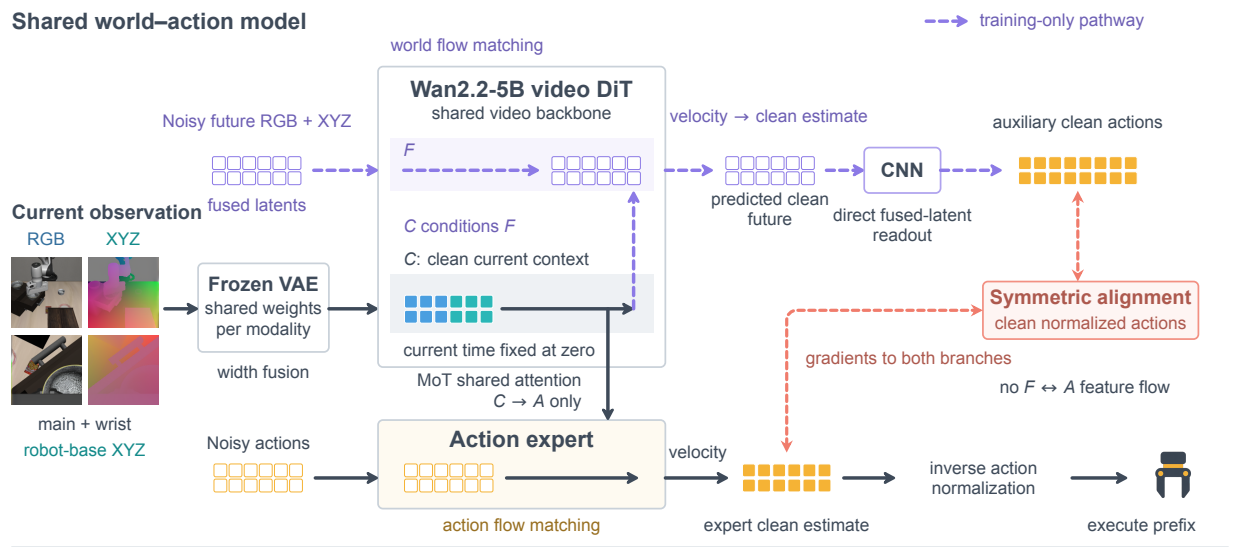

Shared world–action model
training-only pathway
world flow matching
Wan2.2-5B video DiT
shared video backbone
Noisy future RGB + XYZ
fused latents
F
velocity → clean estimate
predicted clean future
CNN
auxiliary clean actions
direct fused-latent readout
Current observation
RGB
XYZ
Frozen VAE
shared weights per modality
width fusion
C conditions F
C: clean current context
current time fixed at zero
MoT shared attention
C → A only
Symmetric alignment
clean normalized actions
gradients to both branches
no F ↔ A feature flow
main + wrist
robot-base XYZ
Noisy actions
Action expert
velocity
inverse action normalization
action flow matching
expert clean estimate
execute prefix
T5 language conditions Wan and the action expert
Online: current K/V prefill, then 10 action-flow updates